

JAYANT CHAUHAN

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate specializing in Machine Learning architectures and low-level systems programming. Demonstrated impact through a highly recognized AI forensics platform (Top 30, India Innovates 2026) and a hackathon-winning video AI pipeline. Contributed to major open-source ecosystems (GNU Compiler Collection, GNU Octave)

EDUCATION

Netaji Subhas University of Technology (NSUT) — New Delhi, India | Expected Graduation: 2028

Bachelor of Technology in Computer Science (99.22 percentile in JEE Main 2024; AIR-10,527 in JEE Advanced 2024)

- Relevant Coursework: Data Structures & Algorithms, OOP (C++), Operating Systems, DBMS, Probability & Statistics, Design & Analysis of Algorithms, Computer Architecture and Organization, Microprocessors and Microcontrollers

OPEN SOURCE EXPERIENCE

GNU Octave — Statistics Package | Open Source Contributor (MATLAB / C++) | Remote | Oct 2025 – Jan 2026

- [PR #345](#) — **stepwisefit Rewrite (Merged)**: Rewrote the stepwise regression backend from scratch, implementing F-test and p-value computation for variable selection loops and Name-Value parsing via pairedArgs convention.
- [PR #347](#) — **dummyvar (Merged)**: Proposed and implemented one-hot encoding for numeric and categorical variables for the first time in the package; handled categories() ordering conventions, NaN-to-zero row semantics, added BISTs.
- [Issue #266](#) — **LinearModel Design Roadmap (Active)**: Authored a 5-phase implementation roadmap for the LinearModel class hierarchy; engaged directly with the package maintainer on class architecture, property schema.

GCC Rust (gccrs) — GNU Compiler Collection | Open Source Contributor | Remote | Jan 2026 – Present

- [Compiler Stabilization](#): Authored and merged 20+ PRs into the GNU Toolchain's Rust frontend, resolving critical Internal Compiler Errors (ICEs) in rust-attributes.cc and adding robust regression tests.
- [AST/HIR Architecture](#): Developed AST and HIR reconstruction logic, creating foundational infrastructure to extend Rust pattern matching support; implemented safety checks across parse, expand, and backend compiler passes.

PROJECTS

SatyaSetu.AI — AI-Powered Digital Forensics & Fraud Detection Platform | **Top 30, India Innovates 2026**

- Designed a full-stack forensic analysis platform combining OCR (Tesseract + PyMuPDF), NER (spaCy), TF-IDF + Sentence-BERT scam classification, XGBoost-based corruption risk prediction into an evidence processing pipeline.
- Trained fraud detection model on 1.8M+ public procurement contracts (Digiwhist Romania 2023), achieving $R^2=0.74$ and RMSE=0.098 on a composite risk indicator target; cross-validated on international datasets for generalization.
- Engineered rule-based fraud signals alongside ML features; exposed predictions via FastAPI with batch analysis and automated PDF forensic report generation.

VividReel — AI-Powered Event Video Summarization | Team Lead | [Hackathon Winner \(Youngest Team\)](#)

- Built an end-to-end CV pipeline (PySceneDetect + OpenCV) that scores clips on focus quality, motion dynamics, and face presence, then assembles cinematic reels via a FastAPI microservice — cutting manual editing time by ~90%.
- Architected multi-stage processing with MoviePy for dynamic assembly, automated transitions, and thematic audio; deployed as a production microservice handling full event footage ingestion to final reel output.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript/TypeScript, SQL, MATLAB/Octave

ML & AI Frameworks: PyTorch, TensorFlow/Keras, Scikit-learn, Transformers, XGBoost, LangChain, LlamaIndex, OpenCV

Data & Tooling: Pandas, NumPy, Jupyter, spaCy, PySceneDetect, MoviePy, Tesseract, PyMuPDF

MLOps & Engineering: FastAPI, Docker, Git/GitHub, CI/CD Pipelines, System Design, Agile/Scrum

LEADERSHIP & VOLUNTEERING

PRAYAS (NSUT) | Core Member

- Mentored underprivileged students in math and science; co-organized a seminar with Microsoft and Salesforce professionals, providing career guidance to 20+ female students.